

Ahmed Rezgui

Paris, France | ahmed.rezgui@insat.ucar.tn | +33 612 742 912 | LinkedIn:ahmed-rezgui | GitHub:ahmedrezgui | Portfolio

Education

National Institute of Applied Sciences and Technology (INSAT)

Tunis, Tunisia

Master's in Software Engineering

Expected Graduation: Sep 2026

- **Concentrations:** Artificial Intelligence / Data Engineering / System Design.
- **Relevant Coursework:** Python, OOP, Data Structures & Algorithms, Statistics, Data Mining, Deep Learning, Image Processing, Business Intelligence, Big Data, Distributed Systems, DevOps.

Experience

BNP Paribas — Inspection Générale

Paris, France

Data Science Intern — Generative AI & AI Agents for Internal Audit

Feb 2026–Aug 2026

- **Designed and built an LLM-as-a-Judge system** to automate quality scoring of internal-audit write-ups (findings, recommendations, sampling) for BNP Paribas Inspection Générale, reducing manual review effort and cutting evaluation time from **hours** per artefact to **minutes**.
- **Engineered a deterministic 3-stage pipeline** (parallel criterion judging → weighted aggregation → synthesis) producing detailed, per-criterion evaluation report with evidence cited from the evaluated artefact.
- **Implemented batch evaluation, caching, Tokens cost tracking and reviewer-override workflows** processing up to ~ **100 audit assignments/run**, reducing redundant LLM calls and associated cost through cache hits on previously scored artefacts.
- Designed and developed an LLM-based advisory system to support senior auditors in audit-recommendation closure decisions; the system produces structured opinions with verbatim citations back to source evidence files (PDF, DOCX, XLSX, ZIP archives).
- Built the system as a multi-agent pipeline in Python using PydanticAI: type-safe Pydantic contracts between layers, versioned prompt files, deterministic inference, and model-level invariants that reject any ungrounded support claim at validation time.
- Deployed with a 100+ test offline suite using pytest, covering parsing different file types, agents, and orchestration, and a per-run JSON audit trail for full decision reproducibility.

Yonnov'IA

Marseille, France

AI Developer

Sep 2025–Feb 2026

- Reduced VRAM training requirements by **93.8%** by using **Quantized Low-Rank Adaptation** to fine-tune an open-source LLM for Enterprise Risk Management, while maintaining comparable performance to full fine-tuning.
- Cut query cost by more than **90%** by implementing **LazyGraph RAG** using **LangChain** and a **locally deployed LLM**.
- Built a full-stack web application using **JavaScript**, **Python**, **React.js** and **FastAPI** to deploy the ERM assistant on a VPS.
- Integrated **Prometheus** and **Grafana** dashboards to monitor the models' and system's performance effectively.

Dydon AI

Zurich, Switzerland

AI/NLP Engineer

Jun 2025–Sep 2025

- Built a **hybrid RAG** microservice in **Python**, **Neo4j**, **FAISS**, **Graphiti** as a shared retrieval layer consumed by internal services.
- Improved retrieval relevance by **20%** using advanced chunking and re-ranking techniques in the RAG pipeline.
- Reduced response latency by **80%** by adding an in-memory cache for previously processed queries.
- Cut the tech lead's monitoring time by **70%** by building an AI-powered dashboard in **JavaScript**, **React.js**, **Python**, **FastAPI** and **Playwright** to automate tracking of team activity on **3 platforms** and auto-generate KPIs/insights for faster decisions.

Projects

StockWise – AI-Powered Stock Market Platform

Jan 2025–May 2025

- Designed and built an end-to-end AI-powered platform for stock market analysis and prediction.
- Developed multi-ticker **LSTM predictors** in **PyTorch** reaching up to **98.3%** accuracy; implemented a **RAG** assistant with **LangGraph** and self-reflective agents for query refinement.
- Automated data collection through an **ETL pipeline** (**Python**, **DVC**, **BeautifulSoup**, **Azure Cloud Storage**).

Skills

- **Programming:** Python, SQL, JavaScript, TypeScript, Java, C++ , C#, HTML/CSS.
- **AI/ML & LLMs:** PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace, LangChain, LangGraph, OpenAI API, QLoRA/PEFT, RAG, AI Agents, LLM-as-a-Judge, OCR, Embeddings & Reranking, MLflow, DVC.
- **Data & Big Data:** Pandas, NumPy, PySpark, Hadoop, Kafka, FAISS, Neo4j, ChromaDB, Pinecone, PostgreSQL, MySQL, MongoDB, SQLite.
- **Engineering & DevOps:** FastAPI, Flask, Gradio, Streamlit, React.js, Next.js, Docker, Kubernetes, Prometheus, Grafana, GitHub Actions, Domino, SonarQube.
- **Languages:** English (fluent), French (fluent), Arabic (native), Spanish (intermediate).